Contents

Bernoulli Trials & Binary Hypothesis Testing

EEE 554 — Lecture #5

Bernoulli Trials

A Bernoulli trial is a random experiment with exactly two possible outcomes, e.g., * $S = \{0, 1\}$ * $S = \{\text{head}, \text{tail}\}$ * $S = \{\text{success}, \text{failure}\}$

It is completely specified by the probability of one outcome, say $P(\text{success}) = p$. The other outcome then has probability $1 - p$.

An n-fold Bernoulli trial consists of a sequence of n independent Bernoulli trials. The outcomes are ordered n-tuples of individual outcomes; e.g., with $n = 3$:

$$S = \{(000), (001), (010), (100), (011), (101), (110), (111)\}$$

Note that S contains 2^n outcomes. Of these: * $\binom{n}{n} = \frac{n!}{n!0!} = 1$ outcome contains n ones. * $\binom{n}{n-1} = \frac{n!}{(n-1)!1!} = n$ outcomes contain $n - 1$ ones. * $\binom{n}{k} = \frac{n!}{(n-k)!k!}$ outcomes contain k ones.

Binomial Coefficient

The quantity

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}, \quad 0 \le k \le n$$

is called a **binomial coefficient** because:

$$(a + b)^n = \binom{n}{0} a^n b^0 + \binom{n}{1} a^{n-1} b^1 + \cdots + \binom{n}{n} a^0 b^n$$

Taking $a = b = 1$ gives:

$$2^n = \sum_{k=0}^{n} \binom{n}{k}$$

These numbers also appear in Pascal's triangle: * $n = 1$: `1 1` * $n = 2$: `1 2 1` * $n = 3$: `1 3 3 1`

If the probability of obtaining a 1 in a single trial is p, the probability of obtaining exactly k ones in n trials is:

$$\binom{n}{k} p^k (1 - p)^{n-k}$$

The probability of obtaining at least k ones in n trials is thus:

$$\sum_{m=k}^{n} \binom{n}{m} p^m (1 - p)^{n-m}$$

Binary Hypothesis Testing

Consider two hypotheses H_0 and H_1 that are mutually exclusive and exhaustive:

$$P(H_0) + P(H_1) = 1$$

These typically represent alternative states: * $H_1 = \{\text{signal present}\}, \quad H_0 = \{\text{signal absent}\}$ * $H_1 = \{\text{transmitted bit } = 1\}, \quad H_0 = \{\text{transmitted bit } = 0\}$

Let M be an event representing an observed measurement or collected data. The decisions are: * $\rightarrow H_1 = \{\text{decide in favor of } H_1\}$ * $\rightarrow H_0 = \{\text{decide in favor of } H_0\}$

Bayesian Decision Rule

The Bayesian Decision Rule decides in favor of H_1 when its conditional probability given the data M is greater than the conditional probability of H_0 given M; i.e.,

$$\rightarrow H_1 = \{P(H_1|M) > P(H_0|M)\}$$

Applying Bayes' rule:

$$\rightarrow H_1 = \left\{ \frac{P(M|H_1)P(H_1)}{P(M)} > \frac{P(M|H_0)P(H_0)}{P(M)} \right\}$$

This simplifies to the likelihood ratio test:

$$\rightarrow H_1 = \left\{ \underbrace{\frac{P(M|H_1)}{P(M|H_0)}}_{\text{Likelihood ratio}} > \underbrace{\frac{P(H_0)}{P(H_1)}}_{\text{Priors}} \right\}$$

Hypothesis Testing with Continuous RVs

Consider the case where the data consists of a realization of one continuous random variable X:

$$M = \{X \text{ takes on a value near } x\} = \{x \leq X < x + \Delta\}$$

Under H_1, X has conditional PDF $f_X(x|H_1)$ and:

$$P(M|H_1) = \int_x^{x+\Delta} f_X(u|H_1)du$$

For small Δ:

$$P(M|H_1) \approx \Delta f_X(x|H_1)$$

Similarly, $P(M|H_0) \approx \Delta f_X(x|H_0)$, and the Bayesian Decision Rule becomes:

$$\rightarrow H_1 = \left\{ \frac{f_X(x|H_1)}{f_X(x|H_0)} > \frac{P(H_0)}{P(H_1)} \right\}$$

Example: Suppose we are given the following conditional PDFs:

$$f_X(x|H_1) = \begin{cases} 4x^3 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f_X(x|H_0) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- **Case 1 (Equal Priors):** With $P(H_0) = P(H_1) = \frac{1}{2}$, the rule is:

$$\to H_1 = \left\{ \frac{4x^3}{1} > 1 \right\} = \left\{ x > \sqrt[3]{\frac{1}{4}} \right\} \approx \{x > 0.63\}$$

- **Case 2 (Unequal Priors):** With H_0 more probable a priori, say $P(H_0) = \frac{3}{4}$ and $P(H_1) = \frac{1}{4}$:

$$\to H_1 = \left\{ \frac{4x^3}{1} > \frac{3/4}{1/4} \right\} = \{4x^3 > 3\} \approx \{x > 0.91\}$$

Metrics of Performance

- **Probability of correct detection (P_d):**

$$P_d = P(\to H_1 \mid H_1) = \int_{\to H_1} f_X(x \mid H_1) dx$$

  It is desirable to have P_d near 1.

- **Probability of false alarm (P_f):**

$$P_f = P(\to H_1 \mid H_0) = \int_{\to H_1} f_X(x \mid H_0) dx$$

  It is desirable to have P_f near 0.

There is usually some compromise necessary between these criteria. For example, one can increase P_d by expanding the decision region for H_1, but this will inherently also increase P_f.

With priors given, one can calculate the overall **Probability of error (P_e)**:

$$\begin{aligned} P_e &= P(\to H_1 \mid H_0)P(H_0) + P(\to H_0 \mid H_1)P(H_1) \\ &= P_f \cdot P(H_0) + (1 - P_d) \cdot P(H_1) \end{aligned}$$